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Module 4 Introduction

Lesson 1: Piecewise polynomials

Lesson 2: Smoothing Splines

Lesson 3: Regularization via Reproducing Kernel Hilbert Spaces

Reading: Regularization via Reproducing Kernel Hilbert Spaces Readings
3h

Video: Regularization and Kernel Functions
8 min

Practice Assignment: Regularization via Reproducing Kernel Hilbert Spaces Quiz
Submitted

Lab: Coding Example
1h

Lab: Coding Exercise
1h

Module 4 Summative Assessment

Module 4 Summary

Your grade: 100%

Next item

Your latest: 100% • Your highest score: 100% • To pass you need at least 80%. We keep your highest score.

Regularization via Reproducing Kernel Hilbert Spaces Quiz

Instructions

Review Learning Objectives

1. What is the primary purpose of regularization in machine learning models? **1 / 1 point**

- To increase the accuracy of the model on the training set.
- To prevent overfitting by adding a penalty term to the model's cost function.
- To transform the features of the model.
- To reduce the computational complexity of the model.

Correct
Correct! Regularization adds a penalty term to reduce overfitting and improve generalization.

2. What is a kernel function in the context of machine learning? **1 / 1 point**

- A function that reduces the dimensionality of the data.
- A function that normalizes the input data.
- A function that calculates the distance between data points.
- A function used to map data into a higher-dimensional space to make it linearly separable.

Correct
Correct! Kernel functions transform data into a higher-dimensional space for better separability.

COACH
Ready to review what you've learned before starting the assignment? I'm here to help.
Help me practice Let's chat

Assignment details

Submitted Nov 16, 1:18 AM IST
Attempts Unlimited
Try again

Your grade
100%
To pass you need at least 80%. We keep your highest score.
View submission **See feedback**